Multiple Deep Proximal Learning for Hyperspectral-Multispectral Image Fusion

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***Abstract*— Fusing low-resolution (LR) hyperspectral image (HSI) with a high-resolution (HR) multispectral image (MSI) could enhance the spatial resolution and quality of HSI. Current deep-learning (DL) HSI-MSI fusion networks have achieved encouraging results, but their performance relies on large number of training images with known degradations consistent with the testing data. The trained DL model may fail on data with unseen degradations during inference. In this study, we propose a multiple deep proximal learning network (MDPro-Net) for HSI-MSI fusion, the unknown spatial–spectral degradations and latent HR HSI can be adaptively inferred. We first propose a joint variational fusion model with both the degradations and HR HSI as to-be-solved variables, which are regularized by multiple deep priors. Then, we optimize the fusion model using quadratic splitting and alternative optimization strategy. The unknown blurring kernel, spectral degradation, and HR HSI are explicitly solved by three deep proximal operators. Through unrolling the solutions into a DL network, we build MDPro-Net, in which the deep proximal operators for degradations and HR HSI are learned in an end-to-end manner. Furthermore, in the deep proximal operator for latent HR HSI, a multiscale transformer is designed to exploit the local and non-local dependencies. Experiments demonstrate that the proposed MDPro-Net is competitive with state-of-the-art fusion methods, in particular, it is robust in inferring the unseen degradations.**

***Index Terms*— Deep, fusion, hyperspectral, multispectral, proximal.**

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I. Introduction

HYPERSPECTRAL image (HSI) acquires the spectral signature of object with narrow bandwidth at each pixel. Utilizing the spatial and spectral information of HSI benefits various computer vision and remote sensing applications, such as land cover classification [1], [2], [3], object detection [4], and tracking [5]. Due to the limited capacity of devices, spatial resolution of HSI is often moderate. The low resolution (LR) of HSI impedes its applicability in scenarios where high resolution (HR) HSI is needed (e.g., fine urban mapping [6] and small target detection [7]).

Multispectral image (MSI), in general, has wider bandwidth and finer spatial resolution than HSI. Fusing with HR MSI captured over the same scene, the resolution of HSI would be effectively enhanced. HSI-MSI fusion is a mainstream approach to enhance the spatial resolution of HSI [8], [9].

In the last decade, a large number of HSI-MSI fusion methods have been proposed. For example, the matrix factorization methods [10], [11], tensor factorization methods [12], [13], [14], [15], [16], and variational fusion methods [17], [18]. Recently, deep learning (DL) has shown its potential in HSI-MSI fusion. By training a DL network representing the mapping among LR HSI, HR MSI, and HR HSI triplets, the HR HSI can be inferred from the LR counterparts [19], [20], [21], [22], [23], [24], [25].

Among the DL HSI-MSI fusion approaches, deep unrolling networks (DUNs) have drawn a lot of research interests [21], [22], [23], [24], [25]. The architecture of unrolling network originates from HSI-MSI fusion model. The fusion model consists of two parts, observation model and different priors [22], [24]. Other than the deep prior, some other priors of latent HR HSI (e.g., low-rankness) can also be integrated in the fusion model [21], [26]. Then, the fusion model is optimized via different optimizers such as proximal gradient decent (PGD) algorithm [22] and alternative direction method of multipliers (ADMMs) [27]. By unrolling the iterative optimization steps into a network, the DL fusion network can be built. Usually, the deep prior of latent HR HSI is implicitly represented by a proximal learning operator [23], [28].

Despite encouraging performance, current DUNs still face some challenges. Most DUNs only involve proximal learning for the latent HR HSI, whereas ignore the proximal learning for spatial–spectral degradations. Without the proximal learning, the ability in predicting the different degradations is limited, thus hinders the networks from adapting to different

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degradations [29]. The DL fusion networks rely on large numbers of training triplets—LR HSIs, HR MSIs, and HR HSIs. A general practice is to obtain the LR HSIs and HR MSIs by degrading existed HR HSIs. We usually assume the degradations (e.g., Gaussian or bicubic downsampling) when we produce the training triplets. However, the real degeneration relationship among LR HSIs, HR MSIs, and HR HSIs is often unknown in reality. Due to the discrepancy between the assumed and real degradations, the learned DL fusion networks may not perform well as it should be. In addition, the degradation varies across sensors and data, so the degradation of testing image may deviate from the training images. Endowing a DL fusion network with the ability to handle various degradations, particularly unseen degradations, is beneficial for real application, but it still lacks study.

To address the above issues, we propose a multiple deep proximal learning network (MDPro-Net) for HSI-MSI fusion. Multiple proximal operators are learned for both latent HR HSI and unknown degradations. We first propose a joint variational fusion model with both degradations and HR HSI as the unknown variables, which are regularized by multiple deep priors. Then, we optimize the fusion model using quadratic splitting and alternative optimization strategy. The unknown blurring kernel, spectral degradation, and HR HSI can be explicitly estimated, and their solutions are represented by three deep proximal operators. To bridge the gap between variational fusion model and DL network, we unroll the solutions into a DL network and build MDPro-Net, in which the deep proximal operators are learned in an end-to-end manner. Furthermore, in the deep proximal operator for HR HSI, a multiscale transformer is designed to exploit the local and non-local dependencies.

The proposed MDPro-Net has two main characteristics.

1) To adaptively predict the unknown degradations during HSI-MSI fusion, we propose a joint variational fusion model. The unknown blurring kernel, spectral degradation, and latent HR HSI are explicitly solved in the fusion model, and their solutions are represented by three deep proximal operators. These deep proximal operators are unrolled into MDPro-Net, which could adaptively infer the degradations and HR HSI. Such ability in inferring degradations is limited in current HSI-MSI fusion networks. On the other hand, MDPro-Net is degradation-adaptive, it does not require that the training and testing data follow the same degradations and can be generalized to sensors with unseen degradations in inference.
2) To exploit the long-range dependencies in the latent HR HSI, inspired by transformers [30], [31], [32], we design a multiscale transformer as deep proximal operator for HR HSI, of which the local and non-local dependencies in the multiscale spaces can be exploited. To alleviate the heavy computation in transformer, the self-attention correlation is calculated along the channels. The experiments demonstrate the effectiveness of multiscale transformer in exploiting the deep prior of latent HR HSI.

The rest of this study is organized as follows. In Section II, we introduce some related works. In Section III, the proposed joint variational fusion model and its optimization are given, the unrolled MDPro-Net is also presented. The experiment results are given and discussed in Section IV. The conclusions are made in Section V.

II. Related Works

A. Model-Based HSI-MSI Fusion

Most of traditional HSI-MSI fusion methods are model-based. For example, with spectral linear mixture assumption, a coupled non-negative matrix factorization model was proposed to decompose the LR HSI and HR MSI into endmembers and abundances; the HR HSI was inferred with the abundance of HR MSI and endmembers of LR HSI [10]. A similar HSI-MSI fusion method based on matrix factorization with sparsity constraint was proposed in Lanaras et al. [11]. Both of the LR HSI and HR MSI are 3-D data, and it is natural to fuse them within tensor framework [12], [13], [14], [15], [16], [33]. In [12], a coupled Tucker tensor decomposition model was proposed, and the LR HSI and HR MSI can be jointly represented as a core tensor and several dictionaries on different modes. The HR HSI was inferred with the core tensor and dictionary on spectral mode of LR HSI and dictionaries on spatial modes of HR MSI. Other tensor factorization models, such as Canonical Polyadic [16], Tensor-train [15], and Tensor-ring models [13], [14], were also used for HSI-MSI fusion.

By exploiting the statistics of latent HR HSI and designing different priors, the HSI-MSI fusion can also be implemented in a variational model. In [17], the subspace sparsity prior of latent HR HSI was proposed for HSI-MSI fusion. Some other priors, such as low-rankness [34], [35], total variation prior [18], and sparsity [34], [36], were also used to regularize the HSI-MSI fusion. However, most of these variational fusion methods assume that the degradations of LR HSI and HR MSI are known, and the fusion performance may deteriorate on data with unknown degradations.

B. Learning-Based HSI-MSI Fusion

Representing the mapping among the LR HSI, HR MSI, and HR HSI triplets, a learned DL network could infer a new HR HSI from the LR HSI and HR MSI pair. In the last decades, many DL networks have been proposed for HSI-MSI fusion [19], [20], [21], [22], [23], [24], [25]. Compared with early networks that only learned brute-force black boxes for the LR-HR mapping [19], [20], recent unrolling fusion networks usually integrate the degradation models [27], [37], [38], [39], [40], [41]. By unrolling the proximal gradient optimization of a fusion model, Dong et al. [22] proposed a model guided fusion network where the degradation was represented by learnable network parameters. Similarly, Xie et al. [21] proposed a low-rank HSI-MSI fusion model and unrolled it into a DL fusion network. In [23], the degradation was implicitly represented by DL network and jointly learned with deep prior in a fusion DL network. In [24], a dual-optimization model for HSI-MSI fusion was unrolled into a DL network. In [25], with a

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lightweight spectral attention embedded to exploit the spectral correlation, an unrolling network was proposed for HSI super-resolution. Via unrolling, some universal fusion networks were proposed, e.g., LR HSI, HR MSI, and panchromatic image were fused in [42], and images with different spectral and spatial resolution were fused in [43]. Most of these DL networks could not fully solve the data fidelity, and the proximal operator is learned only for the latent HR HSI, rather than the degradations. So their ability in inferring various degradations is limited, and they may fail to generalize on data with unseen degradations. How to adapt to different types of degradations within a single learned network still lacks investigation.

Compared with the above supervised DL fusion networks, the unsupervised DL fusion networks require no HR ground-truth HSIs as training labels [44], [45], [46], [47], [48], [49]. A pioneering unsupervised fusion network was proposed in Qu et al. [44], which learned to decompose LR HSI and HR MSI into endmembers and abundances, and then reconstructed the HR HSI with the endmembers of LR HSI and abundances of HR MSI. Similarly, a cross-attention coupled unmixing network with sparsity constraint was proposed for unsupervised HSI-MSI fusion [45]. The HR HSI can also be reconstructed from random noise with deep image prior represented by DL network in an unsupervised way [46]. In [47], a zero-shot fusion method was proposed by generating degraded images using estimated degradation. Xie et al. [49] proposed a positive cycle between fusion and degradation under a probabilistic framework for HSI-MSI fusion. In general, the unsupervised fusion networks need to be trained once when applied to each data.

III. Multiple Deep Proximal Learning Network

A. Observation Model

The observed LR HSI and HR MSI are denoted as $\mathcal{Y} \in \mathbb{R}^{w\times h\times L}$ and $\mathcal{Z} \in \mathbb{R}^{W\times H\times l}$, where w, h, W, and H are the width and height of LR HSI and HR MSI respectively, L and l are the number of their spectral bands. By convention, $w < W$, $h < H$, $l < L$, and $r = W/w = H/h$ is the upscaling factor of HSI-MSI fusion. The aim of HSI-MSI fusion is to estimate the latent HR HSI $\mathcal{X} \in \mathbb{R}^{W\times H\times L}$ by fusing $\mathcal{Y}$ and $\mathcal{Z}$.

The observed LR HSI can be modeled as a spatially degraded version of HR HSI via convolving with a blurring kernel and downsampling, which can be written as

$$\mathcal{Y} = (\mathbf{K} * \mathcal{X}) \downarrow + \mathcal{N}_{\mathcal{Y}} \tag{1}$$

where $\mathbf{K}$ is the blurring kernel, $*$ denotes the convolution in each band of $\mathcal{X}$, and $\downarrow$ is the downsampling in spatial domain. $\mathcal{N}_{\mathcal{Y}}$ is the additive noise or modeling error. The blurring kernel is determined by the point spread functions (PSFs) of imagers. The observed HR MSI can be formulated as a spectrally degraded version of HR HSI as

$$\mathcal{Z} = \mathcal{X} \times_3 \mathbf{R} + \mathcal{N}_{\mathcal{Z}} \tag{2}$$

where $\mathbf{R} \in \mathbb{R}^{l\times L}$ is the spectral degradation matrix, which describes the degeneration relationship between HR HSI and HR MSI. $\mathcal{N}_{\mathcal{Z}}$ is the additive noise or modeling error. $\mathbf{R}$ depends on the spectral response functions (SRFs) of sensors. Both of $\mathbf{K}$ and $\mathbf{R}$ are unknown and need to be estimated.

B. Joint Variational Fusion Model

In HSI-MSI fusion, the blurring kernel $\mathbf{K}$, spectral degradation $\mathbf{R}$, and latent HR HSI $\mathcal{X}$ are unknown and should be jointly estimated. To this end, a joint variational fusion model is proposed as the following objective function:

$$\hat{\mathcal{X}}, \hat{\mathbf{K}}, \hat{\mathbf{R}} = \arg\min_{\mathcal{X},\mathbf{K},\mathbf{R}} \|\mathcal{Y} - (\mathbf{K} * \mathcal{X}) \downarrow\|_F^2 + \|\mathcal{Z} - \mathcal{X}\times_3\mathbf{R}\|_F^2 + \alpha J_{\mathcal{X}}(\mathcal{X}) + \beta J_{\mathbf{K}}(\mathbf{K}) + \gamma J_{\mathbf{R}}(\mathbf{R}) \tag{3}$$

where $\|\cdot\|_F^2$ is the Frobenius norm of tensor. The first two terms are data fidelity terms, which require that the spatial and spectral degraded counterparts of the latent HR HSI $\mathcal{X}$ to be close to the observed LR HSI $\mathcal{Y}$ and HR MSI $\mathcal{Z}$. The terms $J_{\mathcal{X}}(\cdot)$, $J_{\mathbf{K}}(\cdot)$, and $J_{\mathbf{R}}(\cdot)$ represent the priors of latent HR HSI $\mathcal{X}$, the unknown blurring kernel $\mathbf{K}$, and spectral degradation $\mathbf{R}$, respectively. Parameters α, β, and γ are the weights of the trade-off among different priors. These priors provide regularization to overcome the ill-posedness of fusion problem. Many hand-crafted priors, such as sparsity [36] and total variation [18], can be used in HSI-MSI fusion. Due to the flexibility and adaptiveness, the learnable deep prior has achieved encouraging performance in image reconstruction tasks [37], [38]. In this study, we propose to jointly learn multiple deep priors $J_{\text{deep}}(\cdot)$ for the latent HR HSI, blurring kernel, and spectral degradation.

The objective function can be minimized via half-quadratic splitting and alternative optimization. To facilitate the optimization, four auxiliary variables $\mathbf{K}'$, $\mathbf{R}'$, $\mathcal{V}_1$, and $\mathcal{V}_2$ are introduced to decouple the fidelity and prior terms. By optimizing with respect to one variable and fixing others, (3) can be decomposed into the following subproblems:

$$\mathbf{K} = \arg\min_{\mathbf{K}} \|\mathcal{Y} - (\mathbf{K}' * \mathcal{X}) \downarrow\|_F^2 + \beta J_{\text{deep}}(\mathbf{K}) \quad \text{s.t. } \mathbf{K} = \mathbf{K}' \tag{4}$$

$$\mathbf{R} = \arg\min_{\mathbf{R}} \|\mathcal{Z} - \mathcal{X}\times_3\mathbf{R}'\|_F^2 + \gamma J_{\text{deep}}(\mathbf{R}) \quad \text{s.t. } \mathbf{R} = \mathbf{R}' \tag{5}$$

$$\mathcal{X} = \arg\min_{\mathcal{X}} \|\mathcal{Y} - (\mathbf{K} * \mathcal{V}_1) \downarrow\|_F^2 + \|\mathcal{Z} - \mathcal{V}_2\times_3\mathbf{R}\|_F^2 + \alpha J_{\text{deep}}(\mathcal{X}) \quad \text{s.t. } \mathcal{X} = \mathcal{V}_1,\ \mathcal{X} = \mathcal{V}_2. \tag{6}$$

The solutions of $\mathbf{K}$, $\mathbf{R}$, and $\mathcal{X}$ at the sth iterative stage can be represented as the following deep proximal operators:

$$\mathbf{K}^{(s)} = \text{prox}_{(\beta/2\lambda)J}(\mathbf{K}'^{(s)}) = \arg\min_{\mathbf{K}} \frac{1}{2}\|\mathbf{K} - \mathbf{K}'^{(s)}\|_F^2 + \frac{\beta}{2\lambda} J_{\text{deep}}(\mathbf{K}) \tag{7}$$

$$\mathbf{R}^{(s)} = \text{prox}_{(\gamma/2\eta)J}(\mathbf{R}'^{(s)}) = \arg\min_{\mathbf{R}} \frac{1}{2}\|\mathbf{R} - \mathbf{R}'^{(s)}\|_F^2 + \frac{\gamma}{2\eta} J_{\text{deep}}(\mathbf{R}) \tag{8}$$

$$\mathcal{X}^{(s)} = \text{prox}_{[\alpha/2(\mu_1+\mu_2)]J}\left(\frac{\mu_1\mathcal{V}_1^{(s)} + \mu_2\mathcal{V}_2^{(s)}}{\mu_1+\mu_2}\right) = \arg\min_{\mathcal{X}} \frac{1}{2}\left\|\mathcal{X} - \frac{\mu_1\mathcal{V}_1^{(s)} + \mu_2\mathcal{V}_2^{(s)}}{\mu_1+\mu_2}\right\|_F^2 + \frac{\alpha}{2(\mu_1+\mu_2)} J_{\text{deep}}(\mathcal{X}). \tag{9}$$

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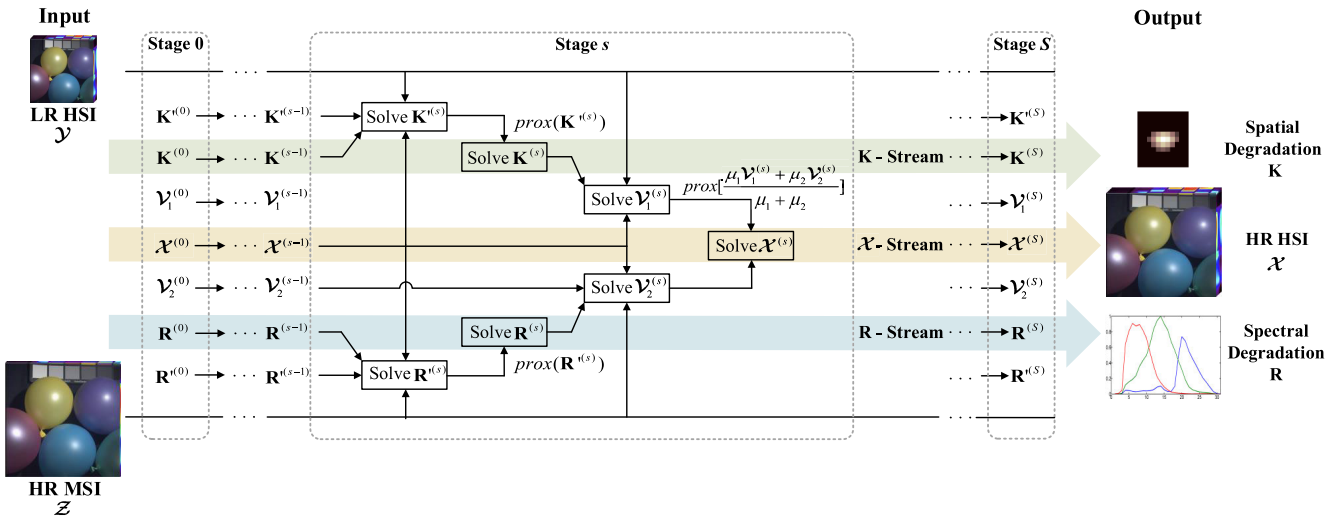


Fig. 1. Overall architecture of MDPro-Net. Each stage corresponds to the iterative solution in Algorithm 1, and the deep proximal operators for solving $\mathcal{X}^{(s)}$, $\mathbf{K}^{(s)}$, and $\mathbf{R}^{(s)}$ are given in Fig. 2. With LR HSI and HR MSI as input, both unknown degradations ($\mathbf{K}$ and $\mathbf{R}$) and HR HSI can be inferred, as illustrated by the three data-streams.

Algorithm 1 Overall Unrolling Process of MDPro-Net

Input: LR HSI $\mathcal{Y}$, HR MSI $\mathcal{Z}$, no. of stages S
Output: HR HSI $\mathcal{X}$, PSF $\mathbf{K}$, SRF $\mathbf{R}$
while $s < S$ **do**
with Eq. (19), solving $\mathbf{K}'^{(s)}$
with Eq. (20), solving $\mathbf{K}^{(s)}$
with Eq. (24), solving $\mathbf{R}'^{(s)}$
with Eq. (25), solving $\mathbf{R}^{(s)}$
with Eq. (28), solving $\mathcal{V}_1^{(s)}$
with Eq. (30), solving $\mathcal{V}_2^{(s)}$
with Eq. (32), solving $\mathcal{X}^{(s)}$
end while

The detailed solutions are given in Appendix. The iterative process of $\mathbf{K}$, $\mathbf{R}$, $\mathcal{X}$, and the auxiliary variables are summarized in Algorithm 1.

C. Network Architecture

By unrolling the iterations in Algorithm 1 into a DL network, we propose the MDPro-Net, as shown in Fig. 1. MDPro-Net consists of multiple stages, corresponding to the iterative process in Algorithm 1. In each stage, there are seven implicit variables (i.e., $\mathbf{K}'^{(s)}$, $\mathbf{K}^{(s)}$, $\mathbf{R}'^{(s)}$, $\mathbf{R}^{(s)}$, $\mathcal{V}_1^{(s)}$, $\mathcal{V}_2^{(s)}$, $\mathcal{X}^{(s)}$) to be solved. The auxiliary variables $\mathbf{K}'^{(s)}$, $\mathbf{R}'^{(s)}$, $\mathcal{V}_1^{(s)}$, and $\mathcal{V}_2^{(s)}$ are solved according to (19), (24), (28), and (30), whereas the HR HSI $\mathcal{X}^{(s)}$, blurring kernel $\mathbf{K}^{(s)}$, and spectral degradation $\mathbf{R}^{(s)}$ are solved by three deep proximal operators, which are described in this section.

1) Deep Proximal Operator for $\mathcal{X}^{(s)}$: The latent HR HSI possesses complex spatial–spectral cross-modality prior that is hard to analytically capture. The proximal operator for $\mathcal{X}^{(s)}$ is designed to learn the prior of latent HR HSI. To exploit the local and long-range dependencies in multiscale spaces of HSI, we design the multiscale transformer proximal operator.

As shown in Fig. 2, the multiscale transformer proximal operator has a U-shaped architecture, which is beneficial for finding out the prior from multiscale spaces [31]. It takes $(\mu_1\mathcal{V}_1^{(s)}+\mu_2\mathcal{V}_2^{(s)})/(\mu_1+\mu_2)$ as input to infer $\mathcal{X}^{(s)}$. At each level of scale, the convolutional layers and transformer block are devoted for local and long-range dependencies, respectively. Taking the starting scale as an example, the convolutional layers project the input into an embedding $\mathcal{F}_0 \in \mathbb{R}^{W\times H\times C}$, where C is the number of feature channels. A transformer block then refines the embedding by exploiting the long-range dependencies. In the transformer block, three 1×1 convolutional layers are first applied to $\mathcal{F}$ and then produce the query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) matrices with reshaping: $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{WH\times C'}$, where C' is the channel of 1×1 convolutional layer. The transformer block refines the feature embedding according to

$$\mathcal{G}_0 = \text{conv}[\text{Reshape}(\mathbf{V}\cdot\text{softmax}(\mathbf{K}^T\mathbf{Q}/\omega))] + \mathcal{F}_0 \quad (10)$$

where $\mathcal{G}_0$ is the output of the transformer block. ω is a learnable scaling parameter controlling the softmax operation. It is noted that the computational complexity of transformer grows quadratically with the spatial size, so we calculate the self-attention correlation along the channels, rather than the spatial domain. Similar to the multihead self-attention, multiple "heads" of query, key, and value matrices can be learned in parallel and then aggregated in the transformer block.

The refined features of the original scale are then gradually downscaled to coarser scales, thus the local and long-range dependencies of multiple scales can be exploited. After reaching the coarsest scale, the features are progressively upscaled to finer scales. Both the down- and up-scaling paths have the same transformer architecture. The down- and up-scaling operations are implemented with pixel-unshuffling and pixel-shuffling, respectively. In order to alleviate gradient vanishing, there are skip connections between the down- and up-scaling paths, and there also exists layer normalization in each transformer block.

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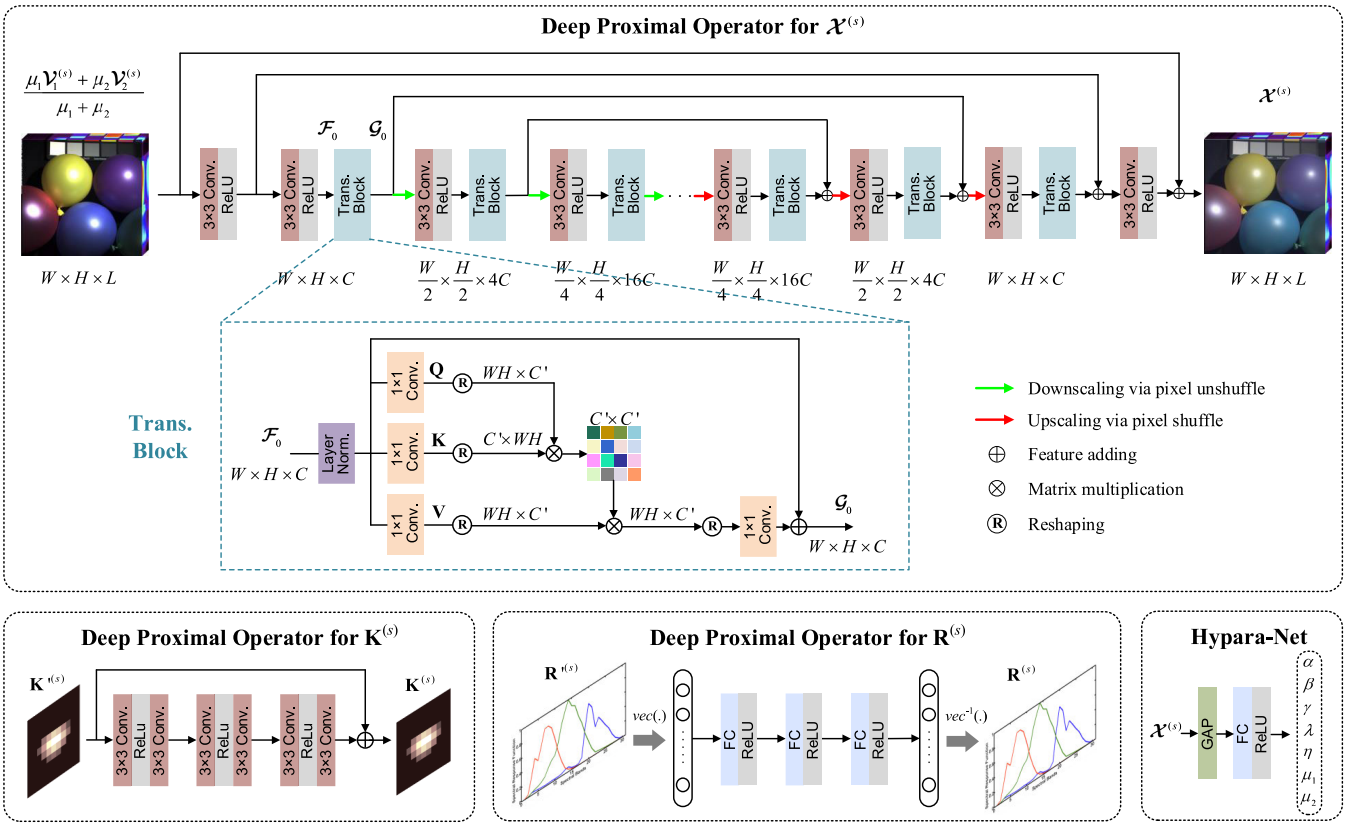


Fig. 2. Three deep proximal operators for solving HR HSI $\mathcal{X}^{(s)}$ and degradations (i.e., $\mathbf{K}^{(s)}$ and $\mathbf{R}^{(s)}$), each of which corresponds to (7)–(9). The Hypara-Net module predicts the hyperparameters.

Finally, in the upscaling path of the original scale, a convolutional layer with L channels follows the transformer to generate the output $\mathcal{X}^{(s)} \in \mathbb{R}^{W\times H\times L}$.

2) Deep Proximal Operator for $\mathbf{K}^{(s)}$: The deep proximal operator of spatial degradation tries to learn the deep prior and then reconstruct $\mathbf{K}^{(s)}$ from $\mathbf{K}'^{(s)}$. Considering that $\mathbf{K}^{(s)}$ is a blurring kernel with a relatively small size, for the sake of simplicity, the deep proximal operator for $\mathbf{K}'^{(s)}$ is designed with residual convolutional network, as shown in Fig. 2. In our implementation, there are three convolutional blocks with 3×3 kernel size and ReLU activation function, which restricts the kernel to be positive. The number of feature channel is set to 16.

3) Deep Proximal Operator for $\mathbf{R}^{(s)}$: The spectral degradation $\mathbf{R}^{(s)}$ is reconstructed from $\mathbf{R}'^{(s)}$ with a deep proximal operator. The spectral degradation $\mathbf{R}^{(s)}$ is a matrix with size $l\times L$, and it can be regarded as a 1-D feature with length lL, so we adopt the fully connected (FC) layers to design the deep proximal operator of $\mathbf{R}^{(s)}$. The input $\mathbf{R}'^{(s)}$ is vectorized and fed to the FC layers. There are two hidden FC layers with 32 neurons per layer. Both input and output layers have lL neurons.

4) Hyperparameters Module: In each stage of MDPro-Net, there is a tiny module learning the hyperparameters (Hypara-Net). Hypara-Net takes $\mathcal{X}^{(s)}$ as input, and then compresses it into a vector via a global average pooling (GAP). There are seven hyperparameters weighting the deep priors and half-quadratic constraints (i.e., $\alpha, \beta, \gamma, \lambda, \eta, \mu_1, \mu_2$), so an FC layer with seven neurons is used to predict the hyperparameters.

D. Training and Implementation Details

All the parameters of MDPro-Net are trained in an end-to-end manner. The L1 norm of reconstruction errors of the latent HR HSI $\mathcal{X}$, the spatial and spectral degradations $\mathbf{K}$ and $\mathbf{R}$ are designed as loss function

$$\text{loss} = \sum_i \|\mathcal{X}_i - \hat{\mathcal{X}}_i\|_1 + \tau_1\|\mathbf{K}_i - \hat{\mathbf{K}}_i\|_1 + \tau_2\|\mathbf{R}_i - \hat{\mathbf{R}}_i\|_1 \tag{11}$$

where $\mathcal{X}_i$, $\mathbf{K}_i$, and $\mathbf{R}_i$ are the ith ground-truths of HR HSI, spatial and spectral degradations, respectively, $\hat{\mathcal{X}}_i$, $\hat{\mathbf{K}}_i$, and $\hat{\mathbf{R}}_i$ are their estimations inferred by MDPro-Net. τ_1 and τ_2 are the weights of degradation losses. By minimizing (11) via back-propagation, the parameters in MDPro-Net (including hyperparameters and learnable parameters in deep proximal operators) can be updated.

In the first stage of MDPro-Net, $\mathcal{X}^{(0)}$, $\mathcal{V}_1^{(0)}$, and $\mathcal{V}_2^{(0)}$ are initialized by upscaling LR HSI $\mathcal{Y}$ with bicubic interpolation. $\mathbf{K}^{(0)}$ is initialized to a Gaussian kernel with standard deviation 1. All elements in $\mathbf{R}^{(0)}$ are initialized to $1/lL$. We set three stages in MDPro-Net. The number of feature channels (i.e., parameter C) in deep proximal operators of $\mathcal{X}^{(s)}$ is set to 64. The loss function is minimized with ADAM optimizer, and the learning rate is 0.0005 and decays by a factor of 0.5 for

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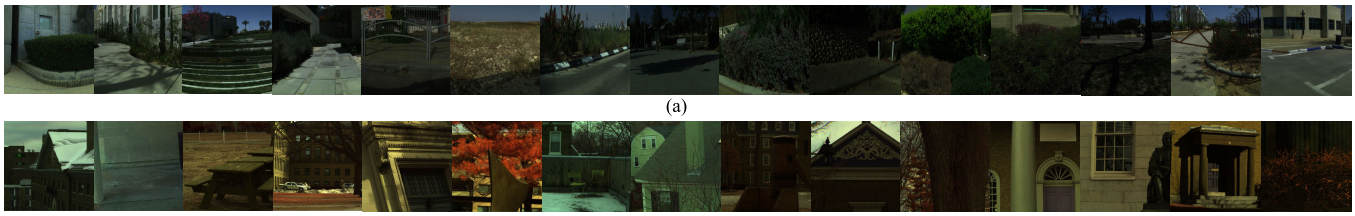


Fig. 3. Fifteen testing images from (a) ICVL and (b) Harvard datasets.

every 10 epochs. The total number of training samples is about 100 000, and the number of training epochs is 100.

IV. Experimental Results

In this section, we compare the performance of MDPro-Net with other state-of-the-art fusion methods on simulated and real remote sensing data. We also analyze the ability in inferring unseen blurring kernel and spectral degradations. The ablation study also demonstrates the effectiveness of the proposed multiscale transformer proximal operators.

A. Experiment Setting

Two widely used datasets are used in the experiments. The first one is the Interdisciplinary Computational Vision Laboratory (ICVL) HSI dataset [51], which was acquired using a Specim PS Kappa DX4 hyperspectral camera. The spatial size of each HSI is 1392 × 1300. There are 31 spectral bands in the spectral range of 400–700 nm. The second dataset is the Harvard dataset [52]. Each HSI in this dataset was collected by a Nuance FX hyperspectral camera with spatial size 1392 × 1040. The Harvard dataset has 31 spectral bands in the spectral range of 420–720 nm.

The original HSIs are used as ground-truth HS HSIs, from which the LR HSIs and HR MSIs are simulated via spatial and spectral degradations. Two libraries containing various PSFs and SRFs are adopted for degrading; one library consists of 300 PSFs with size 11 × 11 [29], and another one has 28 SRFs measured from real imagers [53]. We randomly select PSF and SRF from these two libraries to simulate the LR HSI and HR MSI. There are 15 images taken from each dataset as testing scenes (shown in Fig. 3), and the rest are used as training data. It is noted that there is no overlapping of PSFs and SRFs between the training and testing images.

The fusion performance is evaluated by peak signal-to-noise-ratio (PSNR, dB), structural similarity index measurement (SSIM), Erreur Relative Globale Adimensionnelle de Synthese (ERGAS), spectral angle mapper (SAM), and $Q2^n$ index. PSNR reflects the mean-square error of inferred HSI. SSIM and $Q2^n$ indices measure the similarity between the inferred and ground-truth HR HSIs. ERGAS evaluates the spatial–spectral reconstruction error, and SAM assesses the spectral distortion of inferred HSI [9].

B. Comparison With State-of-the-Art Methods

In this section, we compare the proposed MDPro-Net with six state-of-the-art HSI-MSI fusion methods, they are: coupled nonnegative matrix factorization (CNMF) [10], hyperspectral subspace-based regularized fusion (HySure) [18], fast fusion based on Sylvester equation (FUSE) [50], deep variational fusion network (VaFuNet) [23], deep blind iterative fusion network (DBIN) [19], and model-guided deep convolutional network (MoG-DCN) [22]. The first three are traditional model based fusion methods, and the last three are based on DL networks.

The mean evaluation indices of the 15 testing images are presented in Tables I and II, the best indices are highlighted in bold. It is clear that the proposed MDPro-Net achieves competitive results, and the indices on ICVL and Harvard datasets are better than others. The performance of traditional model-based methods is weaker than the learning-based methods, most of these methods assume the Gaussian PSF in spatial degradation, and thus the non-Gaussian PSFs degrade the performance. Learning-based methods such as VaFuNet [23] and MoG-DCN [22] could learn the degradations from the training data. However, the proximal operator is only learned for the HR HSI instead of degradations, so their ability in adapting to different degradations is limited. The performance may deteriorate if the degradation of testing image deviates from the training data. With multiple proximal learning for both degradations and latent HR HSI, MDPro-Net could explicitly infer the unknown degradations, and has higher adaptability on various degradations, which could achieve satisfactory fusion results on data with unseen degradations.

We give the fusion results on ICVL and Harvard datasets in Figs. 4–7. In addition to the HR HSIs in Figs. 4 and 6, we also show the mean square error maps in Figs. 5 and 7, which reflect the reconstruction error at each pixel. In the results of model-based fusion methods, there are some visible artifacts as shown in Figs. 4(a)–(c) and 6(a)–(c). The learning-based methods perform better than the model-based methods, but compared with MDPro-Net, the compared DL methods still lead to slight artifacts [e.g., Figs. 4(d) and 5(d)] or color distortion [e.g., Fig. 6(f)]. Generally, the reconstruction errors of MDPro-Net are less than others as shown by the mean square error maps.

Table III provides the comparison of training and testing time of different methods. The learning-based methods are implemented on GPU, while the model-based methods are conducted on CPU. Since the model-based methods involve no training, only testing time is compared. Among the learning-based methods, the training time of MDPro-Net is comparable with MOG-DCN, but the testing time is longer, which may result from the multiple deep proximal operators for inferring the degradations and HR HSI.

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TABLE I

EVALUATION INDICES OF DIFFERENT HSI-MSI FUSION METHODS, THE UPSCALING FACTOR IS 4

Dataset	Indices	CNMF [10]	HySure [18]	FUSE [50]	VaFuNet [23]	DBIN [19]	MoG-DCN [22]	MDPro-Net
ICVL	PSNR (dB)	46.0308	44.5704	42.1065	46.7766	43.5182	45.1240	**53.0918**
	SSIM	0.9905	0.9884	0.9738	0.9962	0.9746	0.9961	**0.9981**
	ERGAS	1.5793	2.0983	2.9529	1.4332	3.0305	1.7081	**0.9890**
	SAM	1.7109	2.5378	3.2883	2.0986	4.5680	1.7446	**1.1305**
	$Q2^n$	0.9535	0.9378	0.9254	0.9308	0.9119	0.9168	**0.9631**
Harvard	PSNR (dB)	41.5570	40.1080	40.8237	42.7052	41.2864	40.3954	**43.7843**
	SSIM	0.9701	0.9637	0.9664	0.9773	0.9732	0.9743	**0.9802**
	ERGAS	2.6885	3.4144	3.1174	2.4805	3.1406	3.1052	**2.2727**
	SAM	2.9360	3.6493	3.6340	2.9447	3.1123	3.3630	**2.4612**
	$Q2^n$	0.8801	0.8651	0.8747	0.9034	0.8900	0.8792	**0.9173**

TABLE II

EVALUATION INDICES OF DIFFERENT HSI-MSI FUSION METHODS, THE UPSCALING FACTOR IS 8

Dataset	Indices	CNMF [10]	HySure [18]	FUSE [50]	VaFuNet [23]	DBIN [19]	MoG-DCN [22]	MDPro-Net
ICVL	PSNR (dB)	39.7463	38.5133	37.5201	46.6560	42.1725	43.7214	**48.5777**
	SSIM	0.9668	0.9556	0.9470	0.9926	0.9877	0.9939	**0.9961**
	ERGAS	1.4506	1.8623	2.1024	0.8311	1.5826	1.2113	**0.6638**
	SAM	2.4841	6.7319	6.6314	2.8992	3.8478	2.3129	**1.8802**
	$Q2^n$	0.9183	0.8750	0.8810	0.9421	0.9229	0.9264	**0.9472**
Harvard	PSNR (dB)	37.5103	36.1476	36.0019	40.0286	39.4914	39.8728	**42.0737**
	SSIM	0.9450	0.9331	0.9273	0.9666	0.9669	0.9687	**0.9770**
	ERGAS	4.0448	4.9936	5.0162	4.0477	3.8282	3.2070	**2.7992**
	SAM	3.6352	6.3839	6.3378	3.8264	3.7834	3.2253	**3.0100**
	$Q2^n$	0.8498	0.7989	0.7937	0.8819	0.8728	0.8633	**0.8968**

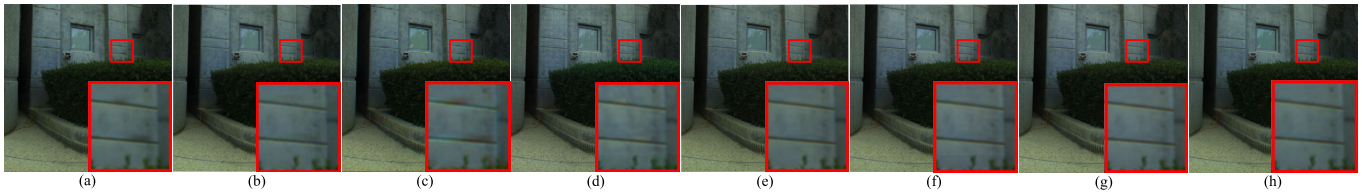


Fig. 4. False color composites (R: 700 nm, G: 600 nm, and B: 500 nm) of different fusion results on ICVL data, the spatial size is 1024 × 1024, the upscaling factor is 8, from left to right: the fusion results of (a) CNMF [10], (b) HySure [18], (c) FUSE [50], (d) DBIN [19], (e) MoG-DCN [22], (f) VaFuNet [23], (g) MDPro-Net, and (h) ground-truth.

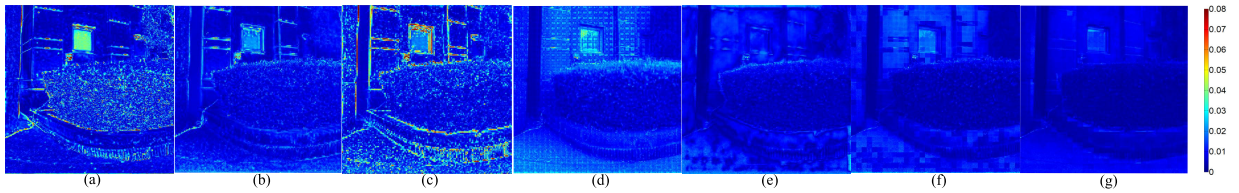


Fig. 5. Mean square error maps of different fusion results in Fig. 4, from left to right: the maps of (a) CNMF [10], (b) HySure [18], (c) FUSE [50], (d) DBIN [19], (e) MoG-DCN [22], and (f) MDPro-Net.

C. Ablation Studies

1) Rationality of Multiscale Transformer: We propose a multiscale transformer proximal operator to exploit the long-range dependencies in the multiple scales of $\boldsymbol{\mathcal{X}}^{(s)}$. With a U-net shape, the proximal operator can be equipped with a transformer in each level. In the default setting of MDPro-Net, we set transformers in both the upscaling and downscaling paths. In Table IV, we show the performance with transformers at different positions. Equipped with transformers in both upscaling (denoted as ↑) and downscaling (denoted as ↓) paths, MDPro-Net achieves the best fusion performance. It can be found that setting transformers in the downscaling path performs better than the upscaling path. Compared with the upscaling path, the downscaling path encodes low-level features, exploiting the long-range dependencies in these features is helpful for image reconstruction. Removing the transformers, the performance of MDPro-Net would drop.

2) Rationality of Learnable Proximal Operators: In the proposed MDPro-Net, the proximal operators are learned in an end-to-end manner together with other network parameters. In this section, taking $\boldsymbol{\mathcal{X}}^{(s)}$ as an example, we replace its deep proximal operator with a fixed denoiser in a plug-and-play manner. Two state-of-the-art DL denoisers, DnCNN [54] and RIDNet [55], are adopted, and the corresponding networks are denoted as MDPro-DnCNN and MDPro-RID. The denoisers are frozen during the training of MDPro-Net. As shown in Table V, since the learnable proximal operator is more flexible

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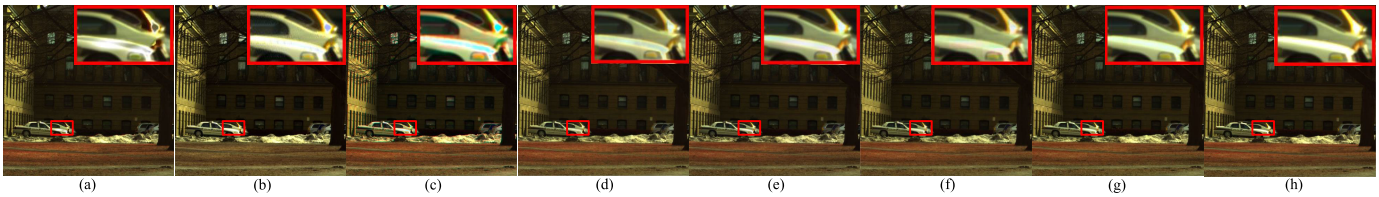


Fig. 6. False color composites (R: 720 nm, G: 620 nm, and B: 520 nm) of different fusion results on Harvard data, the spatial size is 1024 × 1024, the upscaling factor is 8, from left to right: the fusion results of (a) CNMF [10], (b) HySure [18], (c) FUSE [50], (d) DBIN [19], (e) MoG-DCN [22], (f) VaFuNet [23], (g) MDPro-Net, and (h) ground-truth.

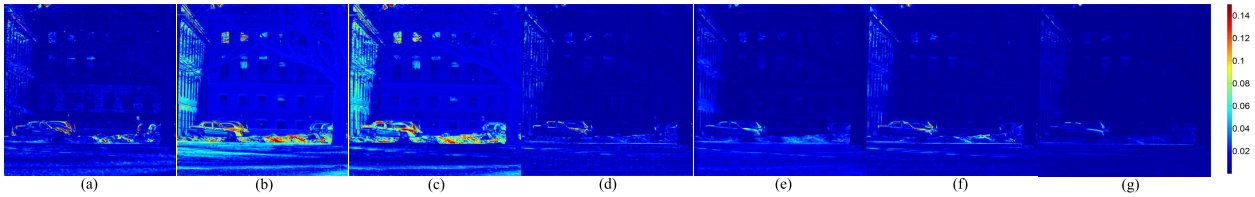


Fig. 7. Mean square error maps of different fusion results in Fig. 6, from left to right: the fusion results of (a) CNMF [10], (b) HySure [18], (c) FUSE [50], (d) DBIN [19], (e) MoG-DCN [22], (f) VaFuNet [23], and (g) MDPro-Net.

TABLE III
COMPARISON OF TRAINING AND TESTING TIME OF DIFFERENT FUSION METHODS (h: HOURS, s: SECONDS)

r	Datasets	CNMF [10]	HySure [18]	FUSE [50]	VaFuNet [23]	DBIN [19]	MoG-DCN [22]	MDPro-Net
×4	ICVL	- / 114.9s	- / 1608.3s	- / 6.7s	∼10h / 5.7s	∼8h / 6.4s	∼20h / 0.7s	∼20h / 26.1s
	Harvard	- / 115.3s	- / 1628.6s	- / 6.6s	∼10h / 5.6s	∼8h / 6.5s	∼20h / 0.8s	∼20h / 25.9s
×8	ICVL	- / 87.0s	- / 1610.0s	- / 6.0s	∼10h / 6.7s	∼8h / 7.1s	∼20h / 0.6s	∼20h / 25.6s
	Harvard	- / 89.5s	- / 1621.3s	- / 6.2s	∼10h / 6.2s	∼8h / 7.0s	∼20h / 0.6s	∼20h / 25.2s

TABLE IV
PSNR (dB) OF MDPRO-NET WITH DIFFERENT TRANSFORMERS IN PROXIMAL OPERATOR FOR $\boldsymbol{\mathcal{X}}^{(s)}$, THE UPSCALING FACTOR IS 4

transformers	ICVL	Harvard
w/ in ↑ and ↓ paths	53.0918	43.7843
w/ in ↑ path	49.4598	41.4250
w/ in ↓ path	50.3830	42.4582
w/o	49.3906	42.0564

TABLE V
PSNR (dB) OF MDPRO-NET WITH LEARNABLE/FIXED PROXIMAL OPERATOR FOR $\boldsymbol{\mathcal{X}}^{(s)}$, THE UPSCALING FACTOR IS 4

	MDPro-DnCNN	MDPro-RID	MDPro-Net (default)
ICVL	41.3470	43.0197	53.0918
Harvard	37.1315	39.1678	43.7843

TABLE VI
PSNR (dB) OF MDPRO-NET WITH DIFFERENT WEIGHTS OF DEGRADATION LOSS, THE UPSCALING FACTOR IS 4

(τ_1, τ_2)	(1, 1)	(1, 0)	(0, 1)	(0, 0)
ICVL	53.0918	49.1797	47.9871	48.8235
Harvard	43.7843	41.4456	42.6051	41.9701

TABLE VII
PSNR (dB) OF MDPRO-NET WITH SELF-ATTENTIONS ON CHANNEL OR SPATIAL DOMAINS, THE UPSCALING FACTOR IS 4

Self-attentions	channel	spatial	channel + spatial
ICVL	53.0918	52.3492	52.3563
Harvard	43.7843	43.8154	43.9022

than the fixed denoiser, MDPro-Net with learnable proximal operator performs better than that with fixed denoiser.

3) Rationality of Degradation Learning: MDPro-Net could explicitly learn the spatial–spectral degradations. The loss function contains three terms, constraining the reconstruction errors of latent HR HSI, blurring kernel, and spectral degradation. We verify the rationality of degradation learning by removing the corresponding terms from the loss function. When the weights τ_1 or τ_2 are set to zero, the loss for blurring kernel or spectral degradation would be removed, and the fusion performance would decrease. If both the spatial and spectral degradation losses are removed, MDPro-Net will get the lowest PSNR. Table VI demonstrates that the degradation learning makes MDPro-Net possible to infer the unknown PSF and SRF, and also benefits its fusion performance.

4) Rationality of Channel Self-Attention: In the proximal operator for $\boldsymbol{\mathcal{X}}^{(s)}$, the dependencies on channel domain are captured by a channel self-attention, and the dependencies on spatial domain are captured by convolutional layers for simplicity. Such architecture helps reduce the computational complexity of network. In this subsection, we compare the self-attentions on channel and spatial domains in Table VII. Putting the self-attention on the spatial domain, the performance of MDPro-Net stabilizes or drops slightly, while the computational would increase. So, we include self-attention only on the channel domain in the proximal operator for $\boldsymbol{\mathcal{X}}^{(s)}$.

D. Some Other Analysis

1) Degradation Robustness Analysis: The proposed MDPro-Net has the ability in inferring unknown degradations.

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TABLE VIII
PSNR (dB) OF MDPRO-NET ON TESTING DATA WITH SEEN DEGRADATIONS

	×4				×8			
	VaFuNet [23]	DBIN [19]	MoG-DCN [22]	MDPro-Net	VaFuNet [23]	DBIN [19]	MoG-DCN [22]	MDPro-Net
ICVL	46.9278	44.1527	46.2757	53.4895	46.8748	43.9301	45.5123	49.7123
Harvard	40.2644	40.0962	41.2950	44.1629	40.2126	39.9375	41.0687	42.9291

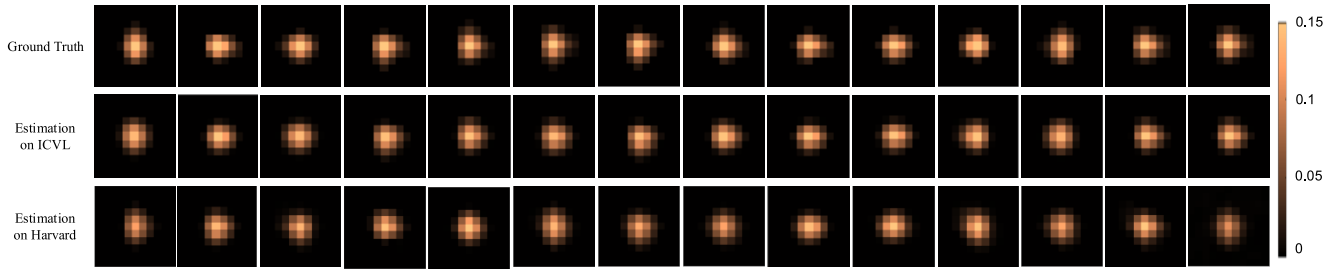


Fig. 8. Fifteen blurring kernels for the testing images in ICVL and Harvard datasets, the first row is the ground-truth, and the second and third rows are the estimated kernels of testing data in ICVL and Harvard datasets, respectively.

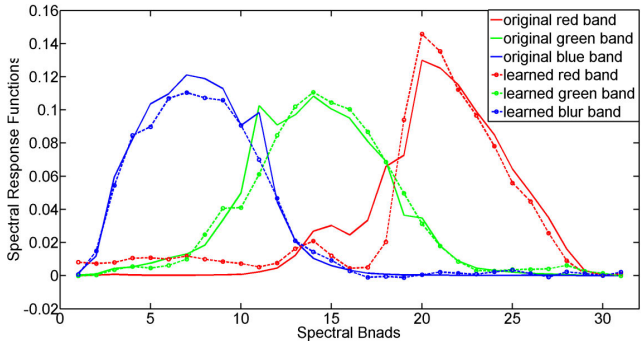


Fig. 9. Visualization of learned and original SRFs.

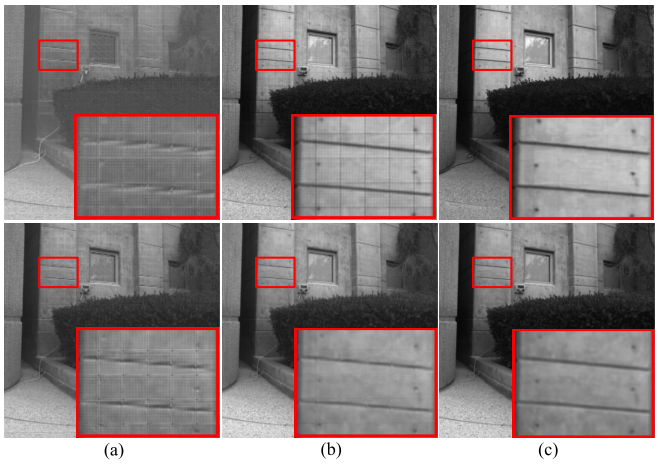


Fig. 10. Inferred HSIs in different stages of MDPro-Net, the upper and bottom rows are 5-th and 25-th channels of (a) $\boldsymbol{\mathcal{X}}^{(1)}$, (b) $\boldsymbol{\mathcal{X}}^{(2)}$, and (c) $\boldsymbol{\mathcal{X}}^{(3)}$.

In Tables I and II, the degradations of training and testing data are totally different, MDPro-Net performs better than other DL fusion networks, which demonstrates its adaptiveness on testing data with unseen degradations. We also analyze their performance on seen degradations in Table VIII, nearly all of the DL fusion networks perform better than the counterparts on data with unseen degradations. Compared with other methods, the proposed MDPro-Net is still competitive on seen degradations, and it demonstrates the robustness on both seen and unseen degradations.

2) Sensitivity Analysis: We also give the sensitivity analysis on the number of stages S of MDPro-Net. In Table IX, it can be found that MDPro-Net with three stages performs better than one stage on both ICVL and Harvard datasets. The larger S means deeper MDPro-Net and better performance. If we further increase the number of stages to five, the PSNR would plateau. So, we fix the number of stages to three in the implementation.

Since the degradations, $\mathbf{K}^{(s)}$ and $\mathbf{R}^{(s)}$, are relatively simple, we design their deep proximal operators with simple network architecture, as described in Section III-C. In this subsection, we give the sensitivity analysis of some network parameters. The performance with more neurons and more layers is better for the deep proximal operator of $\mathbf{R}^{(s)}$. When the number of neurons and layers are larger than 32 and 2, respectively, the performance plateaus, as shown in Tables X and XI. So, we set the number of neurons and layers to 32 and 2. Similarly, the performance also improves with more CNN blocks and more feature channels per layer in the deep proximal operator of $\mathbf{K}^{(s)}$. After the number of CNN blocks and channels increases to 3 and 16, the performance plateaus as well, as shown in Tables XII and XIII. So, we set the number of CNN blocks and channels to 3 and 16.

TABLE IX
PSNR (dB) OF MDPRO-NET WITH DIFFERENT STAGES, THE UPSCALING FACTOR IS 4

	$S=1$	$S=3$	$S=5$
ICVL	50.4015	53.0918	53.1327
Harvard	42.3753	43.7843	43.2312

E. Visualization Analysis

1) Visualization of Learned Spatial-Spectral Degradations: We visualize the learned spatial and spectral degradations (i.e., $\hat{\mathbf{K}}$ and $\hat{\mathbf{R}}$). In Fig. 8, and we give the 15 blurring kernels for the testing images from ICVL and Harvard datasets. The first

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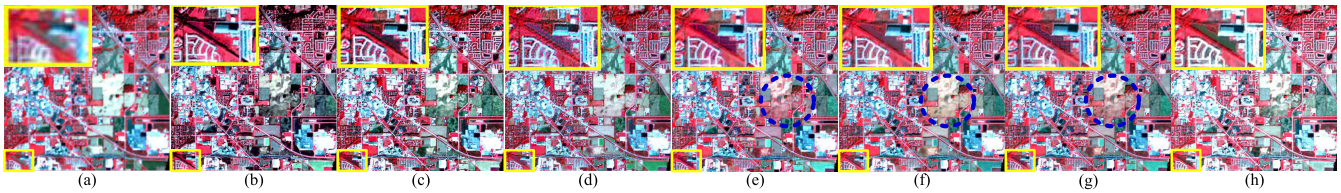


Fig. 11. False color composites (R: 803.3 nm, G: 681.2 nm, and B: 579.5 nm) of different fusion results on real-life spaceborne Hyperion-S2 HSI-MSI fusion, the size of study area is 630 × 630 (a) original Hyperion HSI, (b) result of CNMF [10], (c) result of HySure [18], (d) result of FUSE [50], (e) result of DBIN [19], (f) result of MOG-DCN [22], (g) result of VaFuNet [23], and (h) result of MDPro-Net.

TABLE X

PSNR (dB) OF MDPRO-NET WITH DIFFERENT NEURONS PER LAYER IN PROXIMAL OPERATOR OF $\mathbf{R}^{(s)}$, THE UPSCALING FACTOR IS 4

	16	32	64
ICVL	52.1961	53.0918	52.0227
Harvard	42.4847	43.7843	43.9850

TABLE XI

PSNR (dB) OF MDPRO-NET WITH DIFFERENT LAYERS IN PROXIMAL OPERATOR OF $\mathbf{R}^{(s)}$, THE UPSCALING FACTOR IS 4

	1	2	3
ICVL	51.7925	53.0918	52.8345
Harvard	42.8268	43.7843	44.1458

TABLE XII

PSNR (dB) OF MDPRO-NET WITH DIFFERENT FEATURE CHANNELS IN PROXIMAL OPERATOR OF $\mathbf{K}^{(s)}$, THE UPSCALING FACTOR IS 4

	8	16	32
ICVL	53.0016	53.0918	53.0487
Harvard	42.6704	43.7843	43.9781

TABLE XIII

PSNR (dB) OF MDPRO-NET WITH DIFFERENT CONV. BLOCKS IN PROXIMAL OPERATOR OF $\mathbf{K}^{(s)}$, THE UPSCALING FACTOR IS 4

	2	3	4
ICVL	52.5749	53.0918	53.2739
Harvard	43.3804	43.7843	44.0128

row is the ground-truth; the second and third rows are the estimated blurring kernels of testing data in ICVL and Harvard datasets, respectively. It can be found that MDPro-Net could adaptively estimate different blurring kernels. The learned blurring kernels are close to the ground-truth kernels. Similarly, the spectral degradation can also be learned. In Fig. 9, the learned SRFs approximate the ground-truth SRFs, and the main patterns such as spectral wavelength coverages of different bands can be captured in the learned SRFs.

2) Visualization of Different Stages: Taking the testing image (in Fig. 4) from ICVL dataset as an example, we visualize the evolution of intermediate HSI $\mathcal{X}^{(s)}$ along the stages of MDPro-Net. In Fig. 10, several channels of the inferred HSI $\mathcal{X}^{(s)}$ in different stages of MDPro-Net are presented. As shown in the enlarged regions of the inferred HSI $\mathcal{X}^{(s)}$, there are grid artifacts in the intermediate stages of MDPro-Net, and the artifacts in $\mathcal{X}^{(1)}$ are worse than $\mathcal{X}^{(2)}$. In general, with the evolution of stages, more iterative computation is involved in the fusion results, the grid artifacts are slighter and the fusion results are clearer.

F. Experiment on Real HSI-MSI Data

To demonstrate the practicability of MDPro-Net on real-life data, we apply it to real spaceborne Hyperion-Sentinel data fusion. The HSI was collected by Hyperion imager carried on EO-1 satellite over Lafayette, LA, USA in October 2015. The MSI data was captured by Sentinel-2A (S2) satellite over the same scene in November 2015. The overlapping coverage of the Hyperion and S2 images has, respectively, 2350 × 990 pixels and 7050 × 2970 pixels with spatial resolution 30 and 10 m. The Hyperion HSI was taken in the spectral range of 400–2500 nm. After removing the noisy and water absorption bands, we select 31 bands from Hyperion HSI for the fusion experiment. In S2 MSI, three bands with 10 m resolution are selected for fusion, their center wavelengths are 490, 560, and 665 nm with bandwidths 65, 35, and 30 nm, respectively.

We crop an area with rich textures as testing area. The area represents 210 × 210 Hyperion pixels and 630 × 630 S2 pixels. The training triplets are extracted from the remainder. Since there is no HR HSI ground-truth in real-life data fusion, we train the DL fusion networks on downscaled data, specifically, downscaling the original data and then fusing them, and the original data is treated as ground-truth. Then, we apply the trained networks to the observed HSI and MSI, and the obtained fusion result is an HR HSI with 10 m resolution. Such experimental setting is also adopted in other literatures [21], [23].

The fusion results of different methods are given in Fig. 11. After fusing with S2 MSI, the Hyperion HSI is enhanced and sharpened. Some fusion methods, e.g., FUSE [50] and VaFuNet [23], suffer from visible artifacts in the fusion, as shown in the magnified regions. By comparing the fused HSIs with the original one, it can be found that there exists color distortion in fusion results of CNMF [10] and HySure [18], and the color in the magnified region of MDPro-Net is closer to the original Hyperion HSI. Other methods such as DBIN [19], MOG-DCN [22], and VaFuNet [23] also lead to color distortion, as shown in the blue circle in Fig. 11(e)–(g). Less color distortion implies lower spectral distortion.

V. CONCLUSION

In this study, we propose MDPro-Net for HSI-MSI fusion. Compared with current unrolling fusion networks, there are multiple deep proximal operators in MDPro-Net for inferring both latent HR HSI and unknown degradations. MDPro-Net is guided by the joint variational fusion model regularized by multiple deep priors, in which the unknown blurring kernel,

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spectral degradation, and latent HR HSI are explicitly solved. The iterative solutions are unrolled into the architecture of MDPro-Net. In each stage of MDPro-Net, there are three deep proximal operators representing the unknown blurring kernel, spectral degradation, and latent HR HSI. The experiments demonstrate the effectiveness of HSI-MSI fusion. In this work, we design the multiscale transformer proximal operator to learn the long-range dependencies in multiscale spaces of latent HR HSI. In the future, other deep proximal operators that could learn more priors can be designed to improve the performance.

APPENDIX

Equations (4)–(6) are solved via half-quadratic splitting and alternative optimization strategy. By optimizing with respect to one variable and fixing others, they can further be decomposed into the following subproblems.

APPENDIX A
SOLVING SPATIAL DEGRADATION $\mathbf{K}$

At the sth iterative stage, with half-quadratic penalty, (4) can be rewritten in an unconstrained function

$$\mathbf{K}^{(s)}, \mathbf{K}'^{(s)} = \arg\min_{\mathbf{K},\mathbf{K}'} \|\mathcal{Y} - (\mathbf{K}' * \mathcal{X}^{(s-1)}) \downarrow\|_F^2 + \beta J_{\text{deep}}(\mathbf{K}) + \lambda \|\mathbf{K} - \mathbf{K}'\|_F^2 \quad (12)$$

where λ measures the weight of quadratic penalty. The above function can be further divided into two subproblems, $\mathbf{K}$- and $\mathbf{K}'$-subproblems, via alternative optimization.

1) $\mathbf{K}'$-*Subproblem:* By fixing $\mathbf{K}$, the auxiliary variable $\mathbf{K}'$ can be solved as

$$\mathbf{K}'^{(s)} = \arg\min_{\mathbf{K}'} \|\mathcal{Y} - (\mathbf{K}' * \mathcal{X}^{(s-1)}) \downarrow\|_F^2 + \lambda \|\mathbf{K}^{(s-1)} - \mathbf{K}'\|_F^2. \quad (13)$$

Denote the im2col operator with block size k as U_k, and the circular padding operator with padding size $(k-1)/2$ as $P_{(k-1)/2}$. Then, (13) can be rewritten as

$$\tilde{\mathbf{K}}'^{(s)} = \arg\min_{\tilde{\mathbf{K}}'} \|\mathcal{Y} - \mathfrak{M}_r \mathfrak{X} \tilde{\mathbf{K}}'\|_F^2 + \lambda \|\tilde{\mathbf{K}} - \tilde{\mathbf{K}}'\|_F^2 \quad (14)$$

where $\mathfrak{X} = U_k(P_{(k-1)/2}(\mathcal{X}^{(s-1)}))$, $\mathcal{Y} = U_k(\mathcal{Y})$, $\tilde{\mathbf{K}}$ and $\tilde{\mathbf{K}}'$ are the vectorized versions of $\mathbf{K}^{(s-1)}$ and $\mathbf{K}'$. $\mathfrak{M}_r$ is the matrix representation of downsampling by a factor of r. Representing the downsampling and convolution operators as matrices, it would be more convenient for optimizing (13). By taking the derivative with respect to $\tilde{\mathbf{K}}'$ and making it be zero, the closed-form solution of (14) can be obtained

$$\tilde{\mathbf{K}}'^{(s)} = \left(\mathfrak{X}^T \mathfrak{M}_r^T \mathfrak{M}_r \mathfrak{X} + \lambda \mathbf{I}\right)^{-1} \left(\mathfrak{X}^T \mathfrak{M}_r^T \mathcal{Y} + \lambda \tilde{\mathbf{K}}\right). \quad (15)$$

However, the matrix inverse in (15) is computational heavy and may cause numerical instability issue. Instead of the closed-form solution, we solve (14) via gradient descent method. The iterative solution at the sth stage is

$$\tilde{\mathbf{K}}'^{(s)} = \tilde{\mathbf{K}}'^{(s-1)} + \delta\left[\mathfrak{X}^T \mathfrak{M}_r^T \mathfrak{M}_r \mathfrak{X} \tilde{\mathbf{K}}'^{(s-1)} - \mathfrak{X}^T \mathfrak{M}_r^T \mathcal{Y} + \lambda(\tilde{\mathbf{K}}'^{(s-1)} - \tilde{\mathbf{K}}^{(s-1)})\right] \quad (16)$$

where δ is the step size of gradient descent. It is noted that only one gradient descent step is involved in (16), which is demonstrated with satisfactory performance. Directly calculating $\mathfrak{X}^T \mathfrak{M}_r^T \mathfrak{M}_r \mathfrak{X}$ is time-consuming due to the large size of $\mathfrak{X}$ generated by the im2col operation. Here, we adopt the fast memory-efficient algorithm in [29], which computes $\mathfrak{X}^T \mathfrak{M}_r^T \mathfrak{M}_r \mathfrak{X}$ as

$$\mathfrak{X}^T \mathfrak{M}_r^T \mathfrak{M}_r \mathfrak{X} = R \circ U_k(\mathbf{F}) \quad (17)$$

$$\mathbf{F} = S_r^{-1}(S_r(\mathcal{X}) * (S_r \circ P_{k-1}(\mathcal{X}))) \quad (18)$$

where $S_r(\cdot)$ and $S_r^{-1}(\cdot)$ are pixel shuffle and un-shuffle by a factor of r. $R(\cdot)$ means flipping the rows of a matrix. $\circ$ denotes the composition of multiple operators. With the fast memory-efficient algorithm, $\mathfrak{X}^T \mathfrak{M}_r^T \mathfrak{M}_r \mathfrak{X}$ costs limited computational burden [29]. $\mathbf{K}'^{(s)}$ can be obtained via

$$\mathbf{K}'^{(s)} = \text{vec}^{-1}(\tilde{\mathbf{K}}'^{(s)}) \quad (19)$$

where $\text{vec}^{-1}(\cdot)$ denotes inverse vectorization.

2) $\mathbf{K}$-*Subproblem:* The blurring kernel $\mathbf{K}^{(s)}$ at the sth stage can be obtained from

$$\mathbf{K}^{(s)} = \arg\min_{\mathbf{K}} \frac{1}{2}\|\mathbf{K} - \mathbf{K}'^{(s)}\|_F^2 + \frac{\beta}{2\lambda} J_{\text{deep}}(\mathbf{K}). \quad (20)$$

The above function is equivalent to a denoising problem, which regards the auxiliary variable $\mathbf{K}'^{(s)}$ as the "noisy" version of $\mathbf{K}^{(s)}$. $J_{\text{deep}}(\mathbf{K})$ is the deep prior on blurring kernel to regularize the "denoising" problem. The solution of (20) can be formulated as $\mathbf{K}^{(s)} = \text{prox}_{(\beta/2\lambda)_J}(\mathbf{K}'^{(s)})$, which denotes a deep proximal operator of $\beta J_{\text{deep}}/2\lambda$ at point $\mathbf{K}'^{(s)}$.

APPENDIX B
SOLVING SPECTRAL DEGRADATION $\mathbf{R}$

Similar to (4), with the half-quadratic penalty, at the sth iterative stage, we rewrite (5) into the unconstrained version

$$\mathbf{R}^{(s)}, \mathbf{R}'^{(s)} = \arg\min_{\mathbf{R},\mathbf{R}'} \|\mathcal{Z} - \mathcal{X}^{(s-1)} \times_3 \mathbf{R}'\|_F^2 + \gamma J_{\text{deep}}(\mathbf{R}) + \eta \|\mathbf{R} - \mathbf{R}'\|_F^2 \quad (21)$$

where η is the weight of quadratic penalty. There are two variables in (21), and we solve them using alternative optimization by decomposing it into two subproblems.

1) $\mathbf{R}'$-*Subproblem:* By fixing $\mathbf{R}$, the auxiliary variable $\mathbf{R}'$ can be solved via

$$\mathbf{R}'^{(s)} = \arg\min_{\mathbf{R}'} \|\mathcal{Z} - \mathcal{X}^{(s-1)} \times_3 \mathbf{R}'\|_F^2 + \eta \|\mathbf{R}^{(s-1)} - \mathbf{R}'\|_F^2. \quad (22)$$

Through tensor unfolding operation, the $\mathbf{R}'$-subproblem could also be expressed in matrix format as

$$\mathbf{R}'^{(s)} = \arg\min_{\mathbf{R}'} \left\|\mathcal{Z}_{(3)} - \mathbf{R}' \mathcal{X}_{(3)}^{(s-1)}\right\|_F^2 + \eta \|\mathbf{R}^{(s-1)} - \mathbf{R}'\|_F^2 \quad (23)$$

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where $\mathcal{Z}_{(3)} \in \mathbb{R}^{l\times WH}$ and $\mathcal{X}_{(3)} \in \mathbb{R}^{L\times WH}$ are the tensor unfolding along mode-3. Similar to the $\mathbf{K}'$-subproblem, (23) can also be optimized using gradient descent method. The iterative solution at the sth stage is

$$\mathbf{R}'^{(s)} = \mathbf{R}'^{(s-1)} + \delta\big[-\mathcal{Z}_{(3)}\mathcal{X}_{(3)}^{(s-1)T} + \mathbf{R}'^{(s-1)}\mathcal{X}_{(3)}^{(s-1)} \times \mathcal{X}_{(3)}^{(s-1)T} - \eta\mathbf{R}^{(s-1)} + \eta\mathbf{R}'^{(s-1)}\big]. \tag{24}$$

2) $\mathbf{R}$-*Subproblem:* The spectral degradation at the sth stage is obtained from the $\mathbf{R}$-subproblem

$$\mathbf{R}^{(s)} = \arg\min_{\mathbf{R}} \frac{1}{2}\|\mathbf{R} - \mathbf{R}'^{(s)}\|_F^2 + \frac{\gamma}{2\eta}J_{\text{deep}}(\mathbf{R}). \tag{25}$$

Similar to the $\mathbf{K}$-subproblem, the $\mathbf{R}$-subproblem is equivalent to a "denoising" problem as well. The solution is also a deep proximal operator $\mathbf{R}^{(s)} = \text{prox}_{(\gamma/2\eta)_J}(\mathbf{R}'^{(s)})$.

Appendix C
Solving HR HSI $\mathcal{X}$

After estimating the spatial–spectral degradations, the latent HR HSI $\mathcal{X}$ can be obtained from the objective function

$$\begin{aligned}\mathcal{X}^{(s)}, \mathcal{V}_1^{(s)}, \mathcal{V}_2^{(s)} &= \arg\min_{\mathcal{X},\mathcal{V}_1,\mathcal{V}_2} \|\mathcal{Y} - (\mathbf{K}^{(s)} * \mathcal{V}_1)\downarrow\|_F^2 \\ &\quad + \|\mathcal{Z} - \mathcal{V}_2\times_3\mathbf{R}^{(s)}\|_F^2 + \alpha J_{\text{deep}}(\mathcal{X}) \\ &\quad + \mu_1\|\mathcal{X} - \mathcal{V}_1\|_F^2 + \mu_2\|\mathcal{X} - \mathcal{V}_2\|_F^2 \end{aligned} \tag{26}$$

where μ_1 and μ_2 are two weights of quadratic constraints. Equation (26) has three variables, and it could be decomposed into the following subproblems via alternative optimization.

1) $\mathcal{V}_1$-*Subproblem:* By fixing $\mathcal{V}_2$ and $\mathcal{X}$, the auxiliary variable $\mathcal{V}_1$ can be solved as

$$\mathcal{V}_1^{(s)} = \arg\min_{\mathcal{V}_1} \|\mathcal{Y} - (\mathbf{K}^{(s)} * \mathcal{V}_1)\downarrow\|_F^2 + \mu_1\|\mathcal{X}^{(s-1)} - \mathcal{V}_1\|_F^2. \tag{27}$$

It is more convenient to solving (27) in Fourier domain, in which the closed-form solution exists [29]

$$\mathcal{V}_1{}^{(s)} = \frac{1}{\mu_1}\mathcal{F}^{-1}\left\{\mathfrak{D} - \bar{\mathfrak{K}}^{(s)}\odot_r \frac{(\mathfrak{K}^{(s)}\odot\mathfrak{D})\Downarrow_r}{\mu_1 + (\bar{\mathfrak{K}}^{(s)}\odot\mathfrak{K}^{(s)})\Downarrow_r}\right\} \tag{28}$$

where $\mathfrak{D} = \bar{\mathfrak{K}}^{(s)}\odot\mathcal{F}(\mathcal{Y}\Uparrow_r) + \mu_1\mathcal{F}(\mathcal{X}^{(s-1)})$, $\mathfrak{K}^{(s)} = \mathcal{F}(\mathbf{K}^{(s)})$, and $\bar{\mathfrak{K}}^{(s)} = \bar{\mathcal{F}}(\mathbf{K}^{(s)})$. $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote fast Fourier transformation (FFT) and its inverse transformation, respectively. $\bar{\mathcal{F}}$ is the complex conjugate of $\mathcal{F}$. $\odot$ is the element-wise multiplication, $\odot_r$ means applying the element-wise multiplication to the $r\times r$ distinct blocks. $\Downarrow_r$ is the distinct block downsampling (averaging the $r\times r$ distinct blocks). $\Uparrow_r$ is the upscaling by a factor of r (upsampling and filling entries with zero).

2) $\mathcal{V}_2$-*Subproblem:* By fixing $\mathcal{V}_1$ and $\mathcal{X}$, the $\mathcal{V}_2$-subproblem is

$$\mathcal{V}_2^{(s)} = \arg\min_{\mathcal{V}_2} \|\mathcal{Z} - \mathcal{V}_2\times_3\mathbf{R}^{(s)}\|_F^2 + \mu_2\|\mathcal{X}^{(s-1)} - \mathcal{V}_2\|_F^2. \tag{29}$$

We solve (29) using one-step gradient descent method. The solution at the sth iterative stage of $\mathcal{V}_2$ is

$$\mathcal{V}_2{}^{(s)} = \mathcal{V}_2{}^{(s-1)} + \delta\big[-\mathcal{Z}\times_3\mathbf{R}^{(s)T} + \mathcal{V}_2{}^{(s-1)}\times_3\mathbf{R}^{(s)}\times_3\mathbf{R}^{(s)T} + \mu_2\mathcal{V}_2{}^{(s-1)} - \mu_2\mathcal{X}^{(s-1)}\big]. \tag{30}$$

3) $\mathcal{X}$-*Subproblem:* By fixing $\mathcal{V}_1$ and $\mathcal{V}_2$, the HR HSI $\mathcal{X}$ can be achieved by

$$\mathcal{X}^{(s)} = \arg\min_{\mathcal{X}} \mu_1\big\|\mathcal{X} - \mathcal{V}_1^{(s)}\big\|_F^2 + \mu_2\big\|\mathcal{X} - \mathcal{V}_2^{(s)}\big\|_F^2 + \alpha J_{\text{deep}}(\mathcal{X}). \tag{31}$$

Similar to $\mathbf{K}$- and $\mathbf{R}$-subproblems, the above $\mathcal{X}$-subproblem is also equivalent to a "denoising" problem. Instead of handcrafted prior, we still propose to learn the deep prior $J_{\text{deep}}(\mathcal{X})$ of latent $\mathcal{X}$ as regularization. Equation (31) can be reformulated as

$$\mathcal{X}^{(s)} = \arg\min_{\mathcal{X}} \frac{1}{2}\left\|\mathcal{X} - \frac{\mu_1\mathcal{V}_1^{(s)} + \mu_2\mathcal{V}_2^{(s)}}{\mu_1+\mu_2}\right\|_F^2 + \frac{\alpha}{2(\mu_1+\mu_2)}J_{\text{deep}}(\mathcal{X}). \tag{32}$$

The solution can be expressed as a proximal operator $\mathcal{X}^{(s)} = \text{prox}_{[\alpha/2(\mu_1+\mu_2)]_J}[(\mu_1\mathcal{V}_1^{(s)} + \mu_2\mathcal{V}_2^{(s)})/(\mu_1+\mu_2)]$, which is produced by a DL denoiser network with $(\mu_1\mathcal{V}_1^{(s)} + \mu_2\mathcal{V}_2^{(s)})/(\mu_1+\mu_2)$ as "noisy" input.

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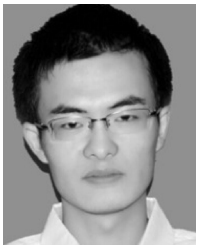
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